

SUB-THEME: FOOD, AGRICULTURE AND HEALTH

Predictive IoT Framework for Early Post-Harvest Spoilage Detection

Group Leader: Adarsh Mishra | **Co-Investigator:** Ranveer

Guide Teacher: Mrs. Sonika Bohra (TGT AI/IT) | **Institution:** Army
Public School, Ahmedabad





The Core Problem: Visual Inspection is Fundamentally Flawed

Too Little, Too Late

In large bulk-storage godowns, managers rely entirely on human eyes to spot rotting produce. By the time brown spots, mushiness, or mold are physically visible, the cellular decay is already severely advanced.

The Chain Reaction

The rot quickly cross-contaminates surrounding healthy stock, causing massive financial losses for warehouse managers and extreme municipal food waste.

The Scientific Methodology: Smell vs. Sight



The Biochemical Reality

When fresh produce spoils, it undergoes anaerobic fermentation and releases invisible volatile organic compounds (VOCs), primarily ethanol gas.

The Time Advantage

These invisible gases are released into the air **28 to 36 hours** before any physical visual decay appears on the skin of the crop.

Our Approach

Our methodology shifts the focus from "*seeing*" the rot with human eyes to "*smelling*" the chemical off-gassing using electronic sensors.

Phase 1 Hardware Node: Our Current Prototype



1

The Edge Node

We engineered a data-acquisition edge node using an Arduino microcontroller.

2

Sensor Integration

The node is equipped with a highly sensitive MQ-3 metal-oxide gas sensor to monitor localized microclimates.

3

Objective Validation

This prototype proves that low-cost hardware can successfully detect invisible ethanol off-gassing in real-time.

4

Alert System

It features a live serial monitor output and a physical threshold backup alarm (Green LED for Healthy, Red LED & Buzzer for Spoilage).

The Final Solution & Future Scope



Phase 1 — Validated ✓

We have successfully validated the hardware's ability to detect localized gas spikes before physical decay occurs.

Phase 2 — In Development

We are currently integrating this live sensory data into a Python-based Artificial Intelligence probability model (Random Forest algorithm).

The Ultimate Solution

The final AI framework will provide godown managers with an automated early-warning smartphone alert, giving them a **1.5-day head start** to isolate at-risk stock and stop massive food waste.